MythosConnect

A voice-first multilingual agent on Gnani Prisma (STT) and Gnani Timbre (TTS)

Aahant Kumar | Delhi Technological University (DTU) | November 2026

2	9 + EN	~3.5 s	11 / 12
Gnani models Prisma + Timbre	Indian languages configured in the app	mean Timbre + Prisma round trip per sentence	benchmark calls succeeded (1 EMPTY_AUDIO failure)

Result in one line: 11 of 12 Timbre-to-Prisma round trips succeeded (mean WER 3.8% on those 11); the one failure, a Telugu EMPTY_AUDIO error, led to a validate-and-retry change in the TTS wrapper.

1. Abstract

MythosConnect lets a user ask a question aloud in one Indian language and hear the answer in another. Gnani Prisma transcribes the speech, a large language model (Gemini) writes a short answer in the target language using a selectable persona and the last three turns of conversation, and Gnani Timbre speaks it back. The system exposes per-stage latency, retries transient failures, and ships with a benchmark (`eval.py`) that uses both Gnani models to measure round-trip transcription accuracy. This report describes the design, the implementation, and the measured results, including the one failure the benchmark exposed.

2. Problem and Use Cases

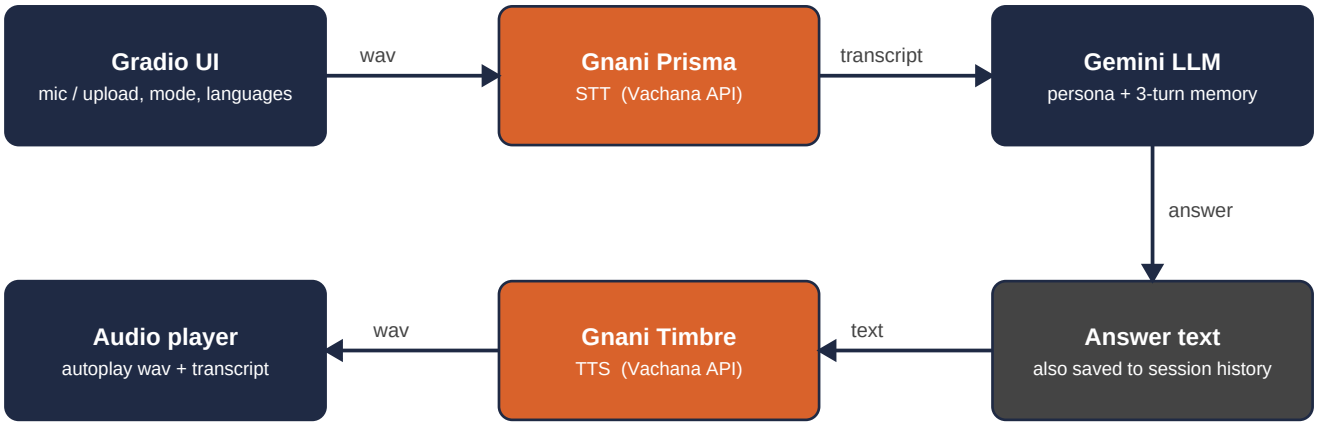
Cultural, educational and civic information is mostly published in English or a few major languages and expects typed input. That excludes low-literacy users, elderly users and people who are most comfortable speaking a regional language. Cross-language access is rarer still: asking in Hindi and hearing the answer in Tamil.

- **Mythology Guide:** heritage and storytelling Q&A; for learners, families, museums and tourism.
- **Student Tutor:** spoken, simple explanations of school topics in the learner's own language.
- **Citizen Helper:** plain-language explanations of services and forms; the prompt instructs the model to say when it is unsure and to point to the official source.
- All three modes share one voice pipeline; a mode is only a system prompt, so new domains cost no new infrastructure.

3. Design Goals

- **Gnani-centred:** both speech directions use Gnani models; the LLM is a swappable reasoning step (Gnani Evon is the planned replacement).
- **Observable:** every stage is timed and every failure names its stage and shows the API's own message.
- **Safe by default:** keys come from environment variables and are whitespace-stripped; nothing secret lives in the repository.
- **Verifiable:** a reproducible benchmark ships with the code and its output is committed as `results.md`.

4. System Architecture



Per-stage timers wrap each arrow; errors short-circuit the pipeline and are shown in the Status box.

4.1 Request lifecycle

Stage	Component	Contract	Failure handling
1 Capture	Gradio audio widget	Mic recording or file upload, saved as wav; recording auto-submits on stop	No audio: status message, no API call
2 Listen	Gnani Prisma STT	POST api.vachana.ai/stt/v3; header X-API-Key-ID; multipart fields audio_file, language_code (optional model)	Retry once on timeout, 429, 5xx; non-200 shown with stage name; empty transcript reported as 'no speech'
3 Think	Gemini (gemini-2.5-flash, configurable)	Prompt = persona + last 3 turns + transcript + output language and style rules	Exception or empty text shown; STT result still displayed
4 Speak	Gnani Timbre TTS	POST api.vachana.ai/api/v1/tts/inference; JSON text, voice, model timbre-v2.5, language, speed, audio_config (48 kHz, mono, 16-bit, wav)	Retry once on transient errors and on empty or truncated audio (< 2 KB); answer text still shown if TTS fails
5 Respond	Gradio outputs	Status + per-stage timings, transcript, answer, autoplay audio, conversation log	Session history updated only after the LLM step succeeds

4.2 Prompt and output design

The prompt requires plain text, at most four short lines, and numbers written as words. These rules are driven by the TTS stage: markdown symbols, bullets and digit strings are read aloud poorly, and short answers keep the Timbre call fast. Memory is a list of (question, answer) pairs held in Gradio session state; only the last three are injected, which bounds prompt size and cost. A "New conversation" button clears the state.

4.3 Configuration surface

Setting	Default	Purpose
STT_MODEL	None (default Prisma model)	Set to "prisma-v3.0" to select Prisma v3
TTS_MODEL / TTS_VOICE	timbre-v2.5 / Nalini	Timbre model and voice
GEMINI_MODEL (env)	gemini-2.5-flash	LLM used for the reasoning step
LANG_MAP	9 languages	Display name to language code (hi-IN, ta-IN, te-IN, ...)
PERSONAS	3 modes	System prompts that define each mode

5. Implementation Notes

The code is a single file of about 225 lines with a clear separation between API wrappers, the agent function and the UI.

Unit	Responsibility
gnani_stt(path, lang)	Uploads audio to Prisma and extracts the transcript from the JSON reply (searches 'transcript' then 'text', including nested objects, so a schema change does not break the app)
gnani_tts(text, lang)	Calls Timbre, accepts either raw audio bytes or base64 JSON, validates size, retries once on empty audio, writes a temp wav file
_post(url, ...)	Shared HTTP helper: 60 s timeout, one retry on network errors, 429 and 5xx, with a short back-off
agent(audio, in, out, mode, history)	Orchestrates the three stages, times each one, builds the prompt, updates history, returns six outputs for the UI
Gradio Blocks UI	Inputs: audio, input language, output language, mode. Outputs: status/latency, transcript, answer, audio, conversation log. Handlers on Ask and on stop_recording
eval.py	Imports the same two Gnani wrappers, so the benchmark exercises exactly the code path the app uses

5.1 Engineering issues found and fixed

Issue	Root cause	Fix
Microphone input never arrived	VS Code's embedded preview browser cannot use the microphone	Launch with inbrowser=True; add a file-upload input as a fallback
Every Gnani call failed at first	Wrong base URL, Bearer auth instead of X-API-Key-ID, wrong field names, invented model names	Rebuilt both calls from the published Vachana API documentation
STT error 'Invalid leading whitespace, reserved character(s), or return character(s) in header value'	A trailing newline was pasted into the API key, which then went into an HTTP header	Strip whitespace when reading both keys from the environment
Intermittent EMPTY_AUDIO from Prisma (benchmark, Telugu)	The wav passed to STT was empty; cause not isolated (Timbre output or transient)	TTS wrapper now validates audio size and retries once, then raises a clear error
Deprecated Gemini SDK, arbitrary model pick	google.generativeai deprecated; model chosen as first in list	Moved to google-genai with a fixed, configurable model

5.2 Security

- API keys are read from environment variables and stripped; `.gitignore` excludes `.env`, audio and editor files, and `.env.example` documents the variable names.
- Keys are never committed; any key exposed during development must be rotated before the repository is made public.
- The Citizen Helper prompt tells the model to admit uncertainty and refer users to official sources, because LLM answers on services can be wrong.

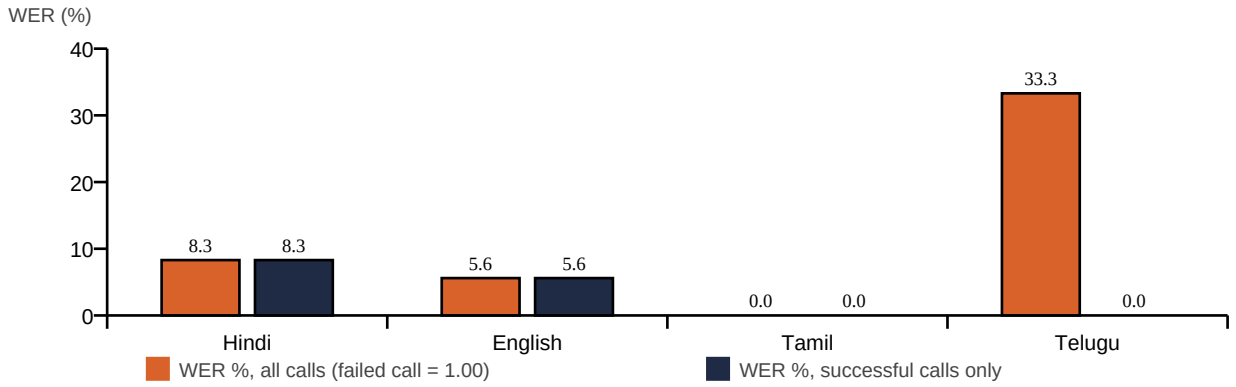
6. Evaluation

6.1 Method

The benchmark closes the loop between the two Gnani models. For each test sentence, Timbre synthesises speech, the audio is sent to Prisma, and the transcript is compared with the original text. Four languages with three short mythology questions each were used (12 sentences). Metric: word error rate, WER = (S + D + I) / N, computed with word-level Levenshtein distance. Normalisation lowercases text and removes ASCII punctuation plus the danda and double danda characters. It deliberately does not use a regex word-character class, because that would strip combining vowel signs in Indic scripts and corrupt the words. Latency is the wall-clock time of the Timbre call plus the Prisma call.

6.2 Results

Language	Sentences	Mean WER, all calls (%)	Mean WER, successful calls (%)	Mean TTS+STT time (s)
Hindi	3	8.3	8.3	3.6
English	3	5.6	5.6	3.4
Tamil	3	0.0	0.0	3.1
Telugu	3	33.3	0.0 (2 of 2)	4.1
All	**12**	**11.8**	**3.8 (11 sentences)**	~3.5 (mean of language means)



6.3 Per-sentence outcomes (romanised; native-script table is in results.md)

Lang	Test sentence	Outcome	WER
Hindi	Ramayan ke lekhak kaun hain	Exact	0.00
Hindi	Hanuman ji ko sankatmochan kyon kaha jata hai	Word split: sankatmochan heard as sankat mochan (1 substitution + 1 insertion over 8 words)	0.25
Hindi	Mahabharat ka yuddh kahan hua tha	Exact (trailing danda removed by normalisation)	0.00
English	Who wrote the Ramayana	Exact (case-insensitive)	0.00
English	Why is Hanuman called Sankat Mochan	One substituted word: mochan heard as mokhan (1 of 6)	0.17
English	Where did the Mahabharata war take place	Exact	0.00
Tamil	Three sentences (Ramayana author; Mahabharata war location; why Vinayagar is worshipped first)	All exact	0.00
Telugu	Ramayanaanni evaru raasaaru; Mahabharata yuddham ekkada jarigindi	Both exact	0.00
Telugu	Vinayakudini mundhuga enduku poojistaaru	Prisma returned 400 EMPTY_AUDIO; scored as WER 1.00	1.00

6.4 Analysis

- **Telugu's 33.3% is one failed call, not a recognition error.** The two Telugu sentences that reached Prisma were transcribed exactly. The benchmark charges a failed call as WER 1.00 on purpose, so reliability is reflected; both figures are reported.
- **The two non-zero successful cases are surface variations**: a compound split into two words in Hindi, and one consonant substitution in a transliterated proper noun in English. Both preserve meaning for the downstream LLM.
- **Latency:** a Timbre plus Prisma round trip took about 3.1 to 4.1 s per sentence on the test machine. Gemini time is not included here; the app reports all three stages separately for each request.

6.5 Threats to validity

- Small sample: 3 sentences per language, so one word moves a language's WER by 12 to 25 points. These are consistency checks, not accuracy claims.
- Synthetic speech from the same vendor, one voice, clean audio. Real human, noisy or accented speech was not tested and is likely harder.
- Short, mythology-domain questions only. The Gemini answer quality and the Hindi-in, Tamil-out experience were not scored quantitatively.
- After the failure above, the TTS wrapper was changed (retry on empty audio). The table reflects the run before that change; a re-run is needed to measure its effect.

7. Limitations and Future Work

Limitation	Planned improvement
Gemini, not a Gnani model, does the reasoning	Swap in Gnani Evon so the full pipeline runs on Gnani models
Whole answer is generated before audio plays	Use Timbre's streaming interfaces to start playback earlier and cut perceived latency
Answers can be wrong or ungrounded	Retrieval over curated source texts; show the sources used
Memory is last-3-turns, in-session only	Summarised long-term memory per user
Benchmark is small and synthetic	Add recorded human clips, report CER as well as WER, more languages and noise conditions
Fixed voice	Voice and speed selectors in the UI

8. Reproducibility

```
pip install -r requirements.txt           # gradio, requests, google-genai
$env:GNANI_API_KEY  = "<key>"            # PowerShell; use export on macOS/Linux
$env:GEMINI_API_KEY = "<key>"
python eval.py     # Timbre -> Prisma benchmark, writes results.md
python app.py      # UI at http://127.0.0.1:7860 (open in Chrome for microphone access)
```

Repository layout: `app.py` (pipeline and UI), `eval.py` (benchmark), `results.md` (latest output), `requirements.txt`, `.env.example`, `.gitignore`, `README.md`.

9. Conclusion

MythosConnect shows that Gnani Prisma and Timbre can be composed with an LLM into a practical voice-first agent for Indian languages in a few hundred lines of code. The measured round trip was accurate on 11 of 12 sentences with about three to four seconds of speech-model time, and the one failure it exposed led directly to a hardening change. The same pipeline serves several use cases by changing only a prompt, and the benchmark makes the claims checkable.